

Growth of small GeO_2 single crystals on a polyvinyl chloride substrate at room temperature using oversaturate aqueous solution

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Abstract

Small GeO_2 single crystal was successfully grown on a polyvinyl chloride (PVC) substrate at room temperature in oversaturate aqueous solution. The oversaturated solution could be prepared at strong acid condition (pH 0) using distilled water, germanium oxide, dilute ammonia, and nitric acid. About $7\ \mu\text{m}$ size of the GeO_2 single crystals with quartz-type structure was obtained by only keeping the PVC plate in the solution for about 10 days when appropriate concentration of the solution was used. The obtained GeO_2 single crystals were confirmed to be constructed by $\{101\}$ and $\{011\}$ surfaces based on powder X-ray diffraction pattern and particle morphology observed by scanning electron microscope.

KEYWORDS

GeO_2 , oversaturated aqueous solution, small single crystal

1 | FOREWORD

As compared to silicon dioxide, germanium dioxide has higher permittivity¹ and more than twice as high piezoelectric constant, while these properties remain up to about $1273\ \text{K}^2$; in addition, germanium dioxide is reported to feature excellent electrical characteristics as a variable resistance memory.³ For this reason, germanium is now being reconsidered as a next-generation material for electronic devices, while chemical characteristics and usage of germanium dioxide are gaining importance in design of advanced electronic devices.⁴ Germanium dioxide can take crystal structure of α -quartz type or cristobalite type, while likely taking amorphous form; though it resembles silicon dioxide, a number of distinctions are known including water solubility and organic polymerization catalytic activity.^{5–7} Germanium dioxide is reported to be water soluble but the solubility value is estimated differently.^{7–10}

We prepared homogeneous aqueous solutions of multiple metal elements including germanium, and developed low-temperature processes to synthesize complex germanates

from such solutions.^{11–14} In the course of development, we discovered that germanium dioxide has abnormal water solubility characteristics. Experimental results obtained for the abnormal solubility of germanium dioxide are overviewed below.

After dispersing germanium dioxide powder in distilled water, diluted ammonia is dropped to produce ammonium germanate soluble in water. Thus, obtained solution is alkaline with pH of about 8. As diluted nitric acid is dropped to the aqueous solution containing ammonium germanate, no deposit is produced until pH decreases below 2. When more nitric acid is added so that a strong acidic solution with pH less than 2 is obtained, deposition occurs; when, after that, diluted ammonia is dropped to increase pH, the deposit does not dissolve until pH reaches about 8.¹⁴ In another experiment, germanium dioxide was dispersed in distilled water but did not dissolve neither when diluted nitric acid was added until pH decreased below 2 nor when diluted ammonia was added until pH increased above 8.¹¹

These results demonstrate that amount of germanium dioxide soluble in water is not determined just by amount of other

coexisting substances. Therefore, we assumed that metastable oversaturated aqueous solution is formed depending on the solution preparation procedure. When oversaturated aqueous solution is brought in contact with heterogeneous solid interface, gradual deposition of solid phase until reaching true solubility may result in growth of single crystals. Dissolution behavior of germanium dioxide via generation of ammonium germanate is reversible; thus, it is highly possible that aqueous solution is obtained in acidic region. In this study, we prepared oversaturated aqueous solutions containing germanium, and attempted at depositing small single crystals of germanium dioxide by using polyvinyl chloride (PVC) as a heterogeneous solid interface. As a result, we discovered that small single crystals of α -quartz type germanium dioxide deposit under certain conditions of oversaturated solution, as reported below.

2 | EXPERIMENTAL METHODS

2.1 | Preparation of oversaturated aqueous solutions

Oversaturated aqueous solutions were prepared under the following protocol. Note that 30 mL ultrapure water and 0.5 g germanium dioxide (Rare Metallic, 99.99%) were mixed into a glass beaker, and then suspension was produced using a magnetic stirrer. In this suspension, pH was approximately 4. Note that 200 mL ammonia water (Nacalai Tesque, 28%) and 200 mL distilled water were mixed into another beaker to obtain diluted ammonia. Then, 400 μ L diluted ammonia was dropped to the germanium dioxide suspension so as to dissolve germanium dioxide so as to obtain a homogeneous aqueous solution. In this homogeneous aqueous solution, pH was approx. 8. Two such oversaturated aqueous solution were prepared, and added with 3 mL and 5 mL, respectively, nitric acid (Nacalai Tesque, specific gravity: 1.38); then the solutions were stirred to obtain oversaturated aqueous solutions. Below, thus obtained oversaturated aqueous solution containing 3 mL and 5 mL nitric acid are referred to as 3HNO solution and 5HNO solution, respectively. Indicated pH value in these solutions was 0.

2.2 | Small single crystal deposition

PVC plates were used as formation substrate for small single crystals of germanium dioxide. A commercially available PVC board was cut into pieces about 2 cm $\times$ 2 cm in size, and then put to ultrasonic cleansing in ethanol. These substrates were immersed in 3HNO and 5HNO solutions, and the beakers were then covered by Parafilm (paraffin film supplied by BEMIS, USA). The PVC substrates were kept in the solutions for 10 days to deposit germanium dioxide. After the dioxide was deposited in 3HNO- and 5HNO solutions,

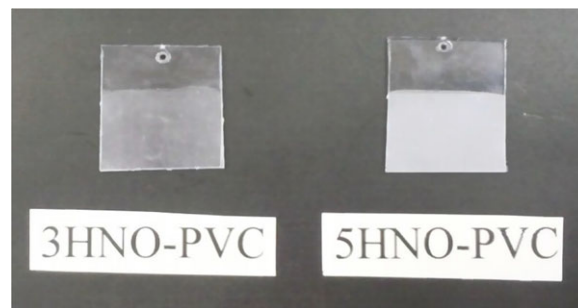


FIGURE 1 Overview of polyvinyl chloride (PVC) substrates on which germanium dioxide microcrystals are deposited in oversaturated aqueous solutions at room temperature [Color figure can be viewed at wileyonlinelibrary.com]

the PVC substrates were cleansed in ethanol and denoted as “3HNO-PVC” and “5HNO-PVC,” respectively. In addition, to confirm acid resistance of the PVC substrates, they were immersed for 10 days in a mixture of 30 mL distilled water, 400 μ L diluted ammonia, and 5 mL nitric acid; after that, the substrates were cleansed in ethanol (below denoted as “Blank-PVC”).

2.3 | Deposit identification

Deposits were identified through analysis of crystalline phase with powder X-ray diffraction (XRD) (MiniFlex, Rigaku Corp., Japan) and morphology observation under scanning electron microscope (Miniscope[®] TM3000, Hitachi High-Technologies Corp., Japan).

3 | RESULTS AND DISCUSSION

Photographs of 3HNO-PVC and 5HNO-PVC are shown in Figure 1. The portions that were in contact with solutions are whitened because of deposits on the surface. Average thickness of deposition on 3HNO-PVC and 5HNO-PVC was 9 μ m and 12 μ m, respectively. Since content of germanium dioxide was same in both oversaturated aqueous solutions, one can assume that amount of deposition depends on the amount of nitric acid.

XRD patterns of 3HNO-PVC, 5HNO-PVC, and Blank-PVC are presented in Figure 2. In XRD patterns of 3HNO-PVC and 5HNO-PVC, the strongest peak was observed at $2\theta \approx 25.9^\circ$, and the only other strong peak was observed at $2\theta \approx 53.3^\circ$. Besides, multiple peaks were confirmed in images enlarged around background. All the peak positions coincided with those of α -quartz type germanium dioxide.¹⁵ Besides, broad peaks in 3HNO-PVC and 5HNO-PVC at $2\theta \approx 15^\circ$ – 25° coincided with the Blank-PVC pattern; that is, these peaks were due to diffraction from PVC plate.

The relatively strong peaks in XRD patterns correspond to 101, 110 peak ($2\theta \approx 25.9^\circ$) and 202, 022 peak ($2\theta \approx 53.3^\circ$);

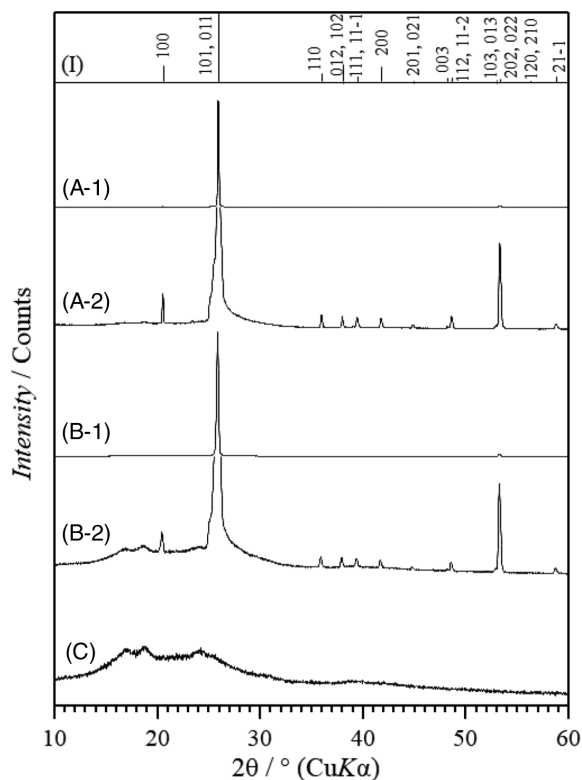


FIGURE 2 X-ray diffraction (XRD) patterns of GeO_2 deposited on PVC plates (The patterns are 3HNO-PVC (A-1), 5HNO-PVC (B-1), and Blank-PVC (C) scaled by strongest peak intensity. The expanded patterns of 3HNO-PVC and 5HNO-PVC are plotted at (A-2) and (B-2), respectively. The reported pattern is also plotted at (I) for comparison.)

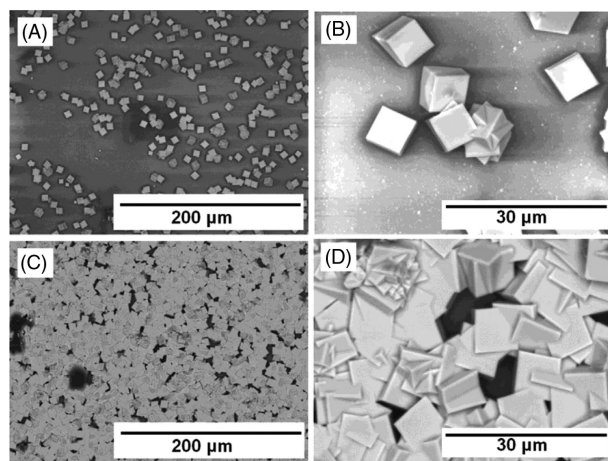


FIGURE 3 Morphology of deposited GeO_2 in 3HNO-PVC ((A) and (B)) and 5HNO-PVC ((C) and (D))

thus, one can conclude that $\{101\}$ or $\{011\}$ planes appeared preferentially on the surface of deposited germanium dioxide crystals.

Low-magnification SEM images in Figure 3A,C indicate that deposition is patchy in 3HNO-PVC, but dense in 5HNO-PVC. In addition, high-magnification images in Figure 4B,D show that the deposits are mostly rhomboid crystals connected

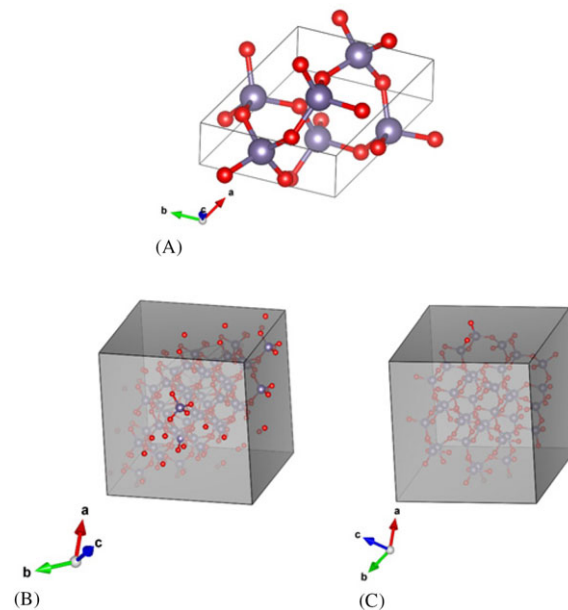


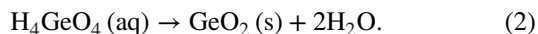
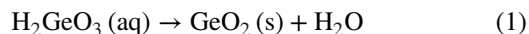
FIGURE 4 (A) Illustration of crystal structure of α -quartz type GeO_2 , (B) single crystal shape constructed by $\{101\}$ surfaces, and (C) single crystal shape constructed by $\{011\}$ surfaces (red and gray balls in (A) present oxygen and germanium atoms, respectively) [Color figure can be viewed at wileyonlinelibrary.com]

to other crystals. Now, α -quartz type belongs to hexagonal crystal systems (see Figure 4A). As distinct from such crystal structure, the shape of single crystals formed by $\{101\}$ or $\{011\}$ planes is rhomboid as shown in Figure 4B,C. Powder XRD analysis applied to the monocrystalline grains in Figure 4B,C shows only diffraction peaks of $h0l$, $0kl$ ($h = k = l$), which agrees with the results of XRD analysis in Figure 1A-1,B-1. Besides, polycrystalline grains composed of multiple connected rhomboid crystals are also observed in Figure 3; thus, one can conclude that crystal planes other than $\{101\}$, $\{011\}$ appear on the surface of connected grains.

In case of 3HNO-PVC, small crystals of germanium dioxide are deposited discretely and many grains have shape specific to single crystals. The single crystal grains in Figure 3B were approximately $7 \mu\text{m}$ in size. As regards synthesis of germanium dioxide particles with similar shape, there are reports about room temperature synthesis using a mixed solution of germanium tetraethoxide, ion exchange water, and L-lysine¹⁶ or room temperature synthesis using germanium dioxide, ammonia, hydrochloric acid, phosphoric acid, and ion exchange water.¹⁷ Those studies dealt with particles of 200–100 nm and $1 \mu\text{m}$, respectively, which is much smaller than the grains handled in this paper. In addition, XRD in those studies were different from this paper; namely, $\{202\}$ and $\{022\}$ diffraction peaks were not relatively strong, which means that single crystal-like structures were not obtained.

Stable compounds in acid aqueous solutions containing germanium are supposed to be neutral compounds such as

H_2GeO_3 ($\text{GeO}(\text{OH})_2$) or H_4GeO_4 ($\text{Ge}(\text{OH})_4$).^{18,19} Both these compounds are supposed to be dissolved in water, and there are no reports about their solid phases. Chemical reactions to generate germanium dioxide from these compounds are shown below.



Here H_2GeO_3 (aq) and H_4GeO_4 (aq) denote respective compounds dissolved in water; GeO_2 (s) denotes solid phase of germanium dioxide. Standard Gibbs energy change in Equations (1) and (2) calculated from reported values¹⁹ and thermodynamic data is -55.3 kJ/mol and -10.8 kJ/mol, respectively, at room temperature. Therefore, the reactions in Equations (1) and (2) are spontaneous. In addition, these absolute values are small, and we assumed that metastable oversaturated solutions can be produced.

In this study, no heat treatment was applied to the crystal deposition substrate, and crystal nucleation sites were not controlled. Therefore, one can expect for growing larger crystals in a shorter time if the substrate surface is provided with crystal formation nuclei. In addition, interaction with the substrate material is unclear, and this issue requires further research.

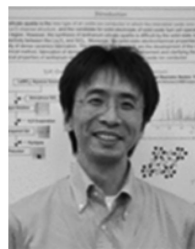
4 | CONCLUSION

Germanium dioxide crystals were deposited on a PVC substrate from aqueous solutions containing germanium dioxide, ammonia, and nitric acid, under atmospheric pressure at room temperature. Amount of thus deposited germanium dioxide depended on the content of nitric acid in aqueous solution. When using solutions with little nitric acid, germanium dioxide deposited discretely, and single crystals about $7 \mu\text{m}$ in size were obtained. These single crystals were confirmed to include {101} and {011} planes.

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